

Scheduling parallel processors: An integer linear programming based heuristic for minimizing setup time

ROBERT T. SUMICHRAS[†] and JOANNA R. BAKER[†]

The problem of scheduling parallel processors in a make-to-stock environment with sequence setup costs is considered. A new algorithm which formulates a series of 0-1 integer subproblems is proposed and contrasted with an earlier formulation (Dearing and Henderson 1982, 1984). Parallels between the subproblem formulations and generalized networks are discussed. The efficiency and quality of the solutions provided were tested using previously published data for a loom assignment problem. The heuristic solution was evaluated against the optimal integer linear programming (ILP) solution, and a rounded linear program (LP) approximation to the optimal solution for several sample problems. Results indicate that the heuristic is efficient, provides near optimal solutions to production planning problems and requires significantly less computing capability than previously reported LP, ILP approaches.

1. Introduction

This paper addresses the special needs of operations which have high and sequence dependent setup times (or costs). Such a situation has been reported in textile (Dearing and Henderson 1982, 1984) and fibreglas (Burch and Sumichrast 1986, Oliff 1982) weaving operations. While most firms are primarily interested in scheduling orders to meet due dates, these operations tend to be more concerned about losing production capacity during the frequent and long setup changes required by the looms. In addition, there may be a concern about the timing of setup changes so that the available setup crew can complete the changes in a timely manner.

Recent papers by Dearing and Henderson (1982, 1984) describe this problem and offer a linear programming model and solution technique. This technique offers good solutions with reasonable computational effort provided a mainframe computer and a good large scale linear programming package are available. The firm which Dearing studied could not implement the scheduling technique because it lacked the required computer package.

In this paper, we propose a simpler scheduling heuristic for minimizing setup costs called MINSET. Because this heuristic requires minimal computational resources, problems of realistic size can be solved on a microcomputer in a few seconds using only a general purpose programming language. Testing the heuristic on a variety of production planning problems shows that it is more computationally efficient than previously reported methods and produces high quality solutions.

2. Problem description

The problem addressed in this study is to schedule a set of identical machines to a set of products so that demand for each product is met and the number of setups is

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[†] Department of Management Science, Virginia Polytechnic Institute and State University, Blacksburg, Virginia 24061, U.S.A.

minimized over the planning horizon. More application specific characteristics are described in the following.

2.1. *Minimizing setup times in a loom operation*

Dearing and Henderson (1982, 1984) have applied math programming to the problem of assigning looms to products over a two quarter planning horizon. Each quarter is divided into 13 weekly time buckets for scheduling purposes. Production capacity was limited by the 460 looms available and demand was expressed in loom-days. The total demand for the five products exceeded the available production during each quarter studied; this imbalance was addressed by allowing management to preference order the products so that demand for the most important products would have a higher chance of being met. No attempt was made to minimize the amount of time devoted to setup changes. Instead, a limit on the time available each week for setup changes smoothed the changes throughout the planning horizon.

Quarter-one data used in this study are reproduced in Table 1 and the setup times are reproduced in Table 2. The setup times are symmetric, meaning that the time it takes to switch a machine from some product i , to another product j , is the same as the time to reverse the switch.

2.2. *The MINSET problem*

Assumptions about the environment in which MINSET is used differ slightly from the Dearing environment. First, MINSET requires a balanced problem—expected production during the horizon must equal the sum of the product demands. This forces management to make choices about which products are really required. Secondly, MINSET is designed to minimize the total setup time throughout the

Table 1. Quarter 1 data for the loom assignment problem.

Product	Initial looms	Management demand in loom-days	Achieved production‡
1	259	2726	2721
2	156	2386	2399
3	45	435	470
4	0	357	36
5	0	357	354

‡ Total production achieved by the Dearing solution (Dearing and Henderson 1982).

Product (i)	Product (j)				
	1	2	3	4	5
1	0	8	24	24	24
2	8	0	24	24	24
3	24	24	0	24	24
4	24	24	24	0	24
5	24	24	24	24	0

Table 2. Setup time for changeovers from product i to product j .

horizon, rather than limiting the number scheduled per week. This characteristic has both good and bad implications.

When more production is required than the weaving operation can produce, minimizing setup time maximizes throughput and therefore increases profit. However, if a particular week requires more setup changes than are possible, then the schedule cannot be used. This apparent limitation can be controlled in practice by dividing the original planning horizon into a series of smaller horizons and solving this series of problems through MINSET. This use of sub-horizons corresponds to Dearing's use of two distinct quarterly solutions rather than one two-quarter horizon. The use of such sub-horizons actually gives management more control of production since it allows demands to be time-phased with more detail. The speed of the heuristic allows several scenarios to be considered.

3. Solution methods

Two methods for assigning looms through mathematical techniques are described in this section. First consider an ILP formulation, the constraint structure of which is similar to the one proposed by Dearing and Henderson. This model is stated below.

Minimize

$$\sum_{i=1}^n \sum_{j=1}^n C_{ij} \sum_{d=1}^H Z_{ijd}$$

subject to

$$X_{k1} + \sum_{j=1}^n Z_{kj1} - \sum_{i=1}^n Z_{ik1} = S_k \quad \text{for } k=1, 2, \dots, n \quad (1)$$

$$X_{kd} + \sum_{j=1}^n Z_{kj d} - \sum_{i=1}^n Z_{ik d} = X_{kd-1} \quad \text{for } k=1, 2, \dots, n \quad \text{and } d=2, 3, \dots, H \quad (2)$$

$$\sum_{d=1}^H X_{kd} = D_k \quad \text{for } k=1, 2, \dots, n, \quad (3)$$

where

W_{id} is the number of machines producing product i on day d ,

Z_{ijd} is the number of machines switched from i to j on day d ,

S_i is the number of machines initially setup to produce product i ,

c_{ij} is the cost to switch a machine from product i to j ,

D_i is the demand for product i (in machine-days),

H is the number of days in the planning horizon, and

n is the number of products to be scheduled,

and all variables are non-negative integers.

The mostly network structure of this formulation makes it possible to display the situation (except for the production constraints) graphically. Figure 1 illustrates this; a three product two day horizon problem has been modelled. One can interpret this model as a flow of machines through product-days, where eqns. (1) and (2) represent conservation of flow on the network. Each horizontal line is associated with one product. Every machine flows into the network, through the lines at the left edge of the graph. The particular arc entered is determined by the product for which the loom is set to produce. In each period of the horizon, all looms move one position to the right, possibly after using a vertical arc to switch between products. The remaining set of constraints (eqn. (3)) forces production in the desired amounts.

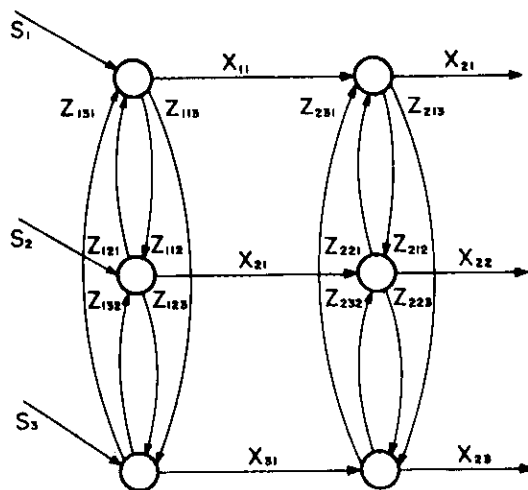


Figure 1.

3.1. The MINSET heuristic

The extensive computations required to solve the Dearing model, rounded LP, even for a feasible solution, have motivated the search for another method of assigning looms in this type of environment. The heuristic MINSET provides a simpler way to find a good solution. This method solves a series of small problems, rather than one large one. As in the Dearing, rounded LP, model, a graphical description can be used to make the model intuitively appealing.

Rather than solving the full problem of scheduling all production, this procedure considers only resources that must be shifted. A series of small 0-1 integer problems will be solved in place of one large ILP. The intuitive benefits of a graphical description as shown in Fig. 1, are still largely maintained, as a series of networks with gains result from the new algorithm. Steps of the algorithm are summarized in the flowchart of Fig. 2 and are explained below.

3.2. Determine capacity needs

First, find the excess capacity or capacity deficiency associated with each of the n products. This is a simple matter since the demand for each product—measured in machine-days—is known and the capacity for production within any setup changes is the product of the planning horizon, and the number of machines initially setup for the product. The amount of excess production associated with a product is then used to determine the greatest number of machines that can leave the product in an optimal solution.

It is clear that in an optimal solution, machines will never be transferred from a product i , and later in the horizon, be transferred back to product i . The maximum number of machines that should be assigned to any product i , at any time is the smallest number that can produce D_i during the horizon and is given by M_i where

$$M_i = \lceil D_i/H \rceil \quad (4)$$

The brackets indicate the smallest integer greater than or equal to the quantity enclosed. Therefore, the maximum number that should leave product i is L_i , where

$$L_i = \lceil S_i - D_i/H \rceil \quad (5)$$

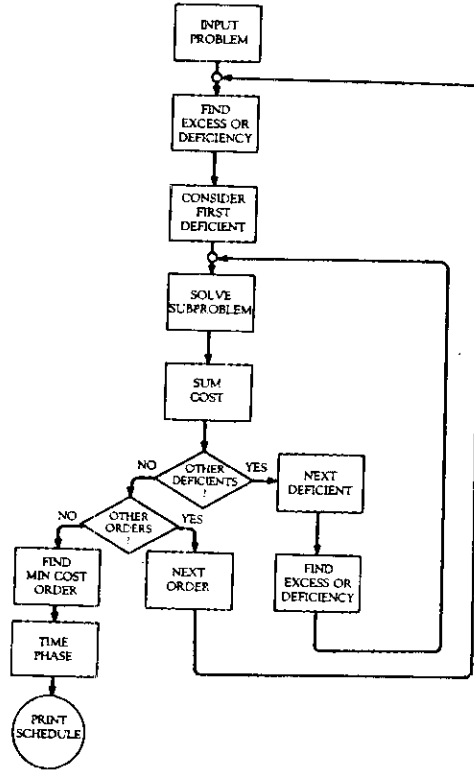


Figure 2.

3.3. Setup ILP sub-problem

The main step is the formulation and solution of a set of 0-1 integer linear programming problems. Each subproblem minimizes the cost of meeting the deficiency of a single product k , from products with surplus capacity and is stated below.

Minimize

$$\sum_i \sum_r c_{ik} Z_{irk} \quad (6)$$

subject to

$$\sum_i \sum_r E_{ir} Z_{irk} - \delta_k = \Delta_k \quad (7)$$

and

$$Z_{irk} \in \{0, 1\}, \delta_k \geq 0 \quad (8)$$

where

- Z_{irk} 1 if the r th machine is switched from product i to product k
0 otherwise
- c_{ik} the cost to switch a machine from product i to product k
- E_{irk} capacity available on the r th machine of product i to be switched to product k

- Δ_k the production deficiency of product k when the problem is formulated
 δ_k the surplus capacity of product k after solution of the current subproblem.

Note that the number of Z_{irk} variables is equal to the number of machines which can leave all products, i.e.

$$\sum_i L_i.$$

3.4. Network interpretation of sub-problems

These formulations can readily be viewed as min-cost-flow networks with gains. The set of source nodes corresponds to the set of machines currently setup for products with excess capacity. Each product i creates L_i nodes. These source nodes have capacities of 1. For each problem formulated, the sink node corresponds to a product with a current deficiency. The sink capacity is the number of machine-days required to fill its production quota. For deficient product k , this is Δ_k . An arc which connects the r th machine capable of leaving the i th product with surplus, has a gain, or multiplier, E_{ir} , equal to the total amount of production which it can supply to the deficient product. This amount can vary from one to H , the number of days in the horizon.

While helpful in visualizing the sub-problem, the network described above is not strictly a min-cost-flow network with gains. The flow is not balanced. Unless there is only one deficient product, the supply will exceed the demand. This is a minor point and can be eliminated through the introduction of a dummy node. More significantly, the requirement that the flows be integer cannot be guaranteed in a generalized network. Therefore, no attempt has been made to exploit the special structure of this problem in finding its solution. The language associated with this network model will, however, continue to be used to describe the problem.

3.5. Obtaining a solution

To implement the main step, an arbitrary order of the deficient products is chosen. The initial excess capacity is shipped to at least fill its demand, at the lowest setup cost. Note that in general, this demand will be overfilled, resulting in a new product with a surplus of capacity. The excess or deficiency associated with each product is now recalculated and a new ILP to supply the next product with a deficiency is solved. The total cost to supply all deficiencies is recorded. This set of steps is now repeated for each possible order of the deficient products and the minimum cost of all orders is chosen.

The chosen solution must finally be converted to a time-phased set of setup changes. Consider the solution to the first sub-problem. For each arc with flow (i.e. $Z_{irk}=1$) determine the day, d_{ir} , to transfer the machine as, $d_{ir}=H-E_{ir}+1$. Time-phasing of subsequent sub-problems is identical.

The lines which connect the supply nodes to the deficient node indicate the amount of capacity (in brackets) and the cost to make the assignment (in parenthesis). A well known algorithm (Sumichrast 1986, Syslo *et al.* 1983) is used to pick which transfers to make. For the example, the cheapest way to at least meet product 5's deficiency is to use one loom from product 1, one from product 2 and one from product 3. This choice is indicated by darkening the associated arcs in the diagram.

A new set of deficiencies and surplus is found based on these assignments, and the reassignment is repeated for each product which remains deficient. Finally, this process is repeated for all orderings of the deficient products and the lowest cost, complete assignment is used.

3.6. Computational resources required

A solution method must require no more computational resources than are available at the installation site if it is to have an impact on operations. Dearing's LP solution method was designed to run on a mainframe computer and required the use of a specialized package (MPSX) to accomplish the main computations. Using an IBM 3033 computer, the main computations were completed in 16 CPU seconds for an example problem with five products and two 13 week horizons. A 10 product problem required two minutes of CPU time. While these computation times are not prohibitive to a user with appropriate computer resources, Dearing's solution technique was not implemented because of a lack of appropriate software.

The MINSET technique requires minimal computer resources. Problems with 6 products are solvable in approximately 15 seconds. The heuristic has been programmed in PASCAL for a microcomputer with at least 64 K RAM. No special package is required for computations and therefore almost any company with this type of problem should find implementation possible.

4. Results

Several tests of MINSET are presented in this section. Comparisons are made to the LP technique using data from the Dearing study. In addition, random problems are generated of a size small enough to permit an optimal integer solution to be found so that some absolute measure of the performance of both the Dearing LP technique and the MINSET heuristic is available.

The MINSET heuristic was used to find loom assignments for the first quarter Dearing data. The five product, 13 week problem was re-solved using the initial loom assignments shown in Table 1. Actual production achieved by the Dearing solution was used as demand. In this way, production capacity and total demand were balanced as required by MINSET. The production plan obtained from MINSET is shown in Table 3. The total setup time is 1234 hours, which is almost a 39% reduction of the solution reported by Dearing and Henderson.

Table 3. Production plan for 5 product, 13 week problem; total setup time = 1234 hours.

Product	Initial looms setup	Demand (loom-days)	Week	Looms switched		
				from prod.	to prod.	Number
1	259	2726	1	1	2	28
			1	1	4	21
			10	1	5	1
2	156	2386	8	2	4	1
3	45	435	1	3	4	6
			1	3	5	5
			7	3	5	1
4	0	357	—	—	—	—
5	0	357	—	—	—	—

If changeover capacity is limited, as assumed in the Dearing study, then this solution probably cannot be used. Virtually all of the setup changes have been scheduled for the first week. While MINSET does not have the capacity to explicitly limit the number of setup changes during a particular week, the demands can be time-phased through the use of sub-horizons so that the need for setup time is smoothed throughout the horizon. This technique is demonstrated below.

The 13 week planning horizon is broken into two 4 week and one 5 week partial horizons. These smaller problems are solved successively, so that the initial setup for each subsequent horizon corresponds to the final loom assignments of the previous solution. The total setup time for the set of sub-horizon problems is 1488 hours, which is still a 14% decrease over the solution reported by Dearing. The MINSET solution is shown in Table 4.

A second problem was solved using the actual demands. A sixth product was created and the planning horizon extended to 14 weeks to provide sufficient capacity. Loom switches to the sixth, or dummy, product represented looms which were turned off; hence, no setup time was incurred for such changes. Demand for the dummy product was equal to the difference between total real demand and total capacity in 14 weeks. The production plan is shown in Table 5. Note that the total number of hours needed for all switches is 1370, a 26% decrease from the previously reported result.

4.1. Random problems

Seven additional randomly generated problems were solved through MINSET. Dearing's rounded LP technique, and through integer linear programming (ILP). ILP provides an absolute performance measure. However, obtaining the optimal integer solution is very inefficient computationally and this severely limited the problem size and the number of comparisons made. Several minutes were required to

Product	Initial looms setup	Demand (loom-days)	Week	Looms switched		Number
				from prod.	to prod.	
1	259	2726	1	1	2	20
			1	1	4	1
			3	1	4	1
			5	1	2	7
			5	1	5	15
			7	1	5	1
			9	1	2	7
			9	1	5	22
			10	1	2	1
			—	—	—	—
2	156	2386	—	—	—	—
3	45	435	14	3	4	1
			1	—	5	5
			4	—	5	1
			5	—	5	3
4	0	357	—	—	—	—
5	0	357	—	—	—	—

Table 4. Production plan for broken horizon 5 product, 13 week problem; total setup time = 1488 hours.

Product	Initial looms setup	Demand (loom-days)	Week	Looms switched		
				from prod.	to prod.	Number
1	259	2726	1	1	2	15
			1	1	4	26
			1	1	5	23
			11	1	6	1
2	156	2386	7	2	5	1
3	45	435	1	3	6	11
4	0	357	1	4	6	2
			8	4	6	1
5	0	357	1	5	6	14
6 (dummy)	0	281	—	6	—	—

Table 5. Production plan for the 6 product, 14 week problem; total setup time = 1470 hours.

solve even the smallest problem, using an IBM 3084. The largest problem tried required approximately 30 minutes of CPU time. As reported in Table 6, MINSET compared favourably with the rounded LP technique.

5. Conclusions and summary

An efficient microcomputer based heuristic for scheduling looms has been presented and compared to an earlier heuristic solution technique and to optimal solutions. This heuristic performed well, based on several criteria. First, if the goal is to schedule looms so that the smallest amount of setup time is used, MINSET produces schedules of at least the same quality as the LP with rounding technique. This was tested with both published data and with new randomly generated problems. The set of random problems are especially important because they do not explicitly limit the number of changes per week and because they were small enough to be solved through ILP for an optimal solution.

A limitation of MINSET is that the number of setup changes cannot be limited per period. A method was presented for minimizing this potential problem. When a schedule requires too many changes during a single time bucket, the planning horizon can be broken into several sub-horizons, and demand time-phased over these smaller time intervals. This provides the planner with more flexibility. Although

Products	Horizon (weeks)	Setup time in hours		
		Minset	Rounded LP	ILP
3	4	285	250	250
4	4	300	240	240
4	5	374	402	374
4	5	244	277	257
5	5	165	368	165
6	5	574	577	523
6	5	479	495	403

Table 6. Comparison of solution methods.

some increase in the total setup time is the inevitable cost of this increased flexibility and smoothed setup change schedule.

This study also pointed out some general guidelines for scheduling an operation with the cost and demand structure considered. If the data reproduced from the Dearing study is representative of the industry, then management should consider increasing their ability to change loom setups during a time period. Allowing more setup changes to be completed during a single period significantly reduced the total setup time required. Dearing's data indicates that more production was required than was available, so it can be concluded that this lack of setup capacity limits the sales of this operation and the cost of lost production can be very significant. Increasing the capacity to change loom setups might be accomplished by either hiring more setup personnel or by training production workers to change the setups on their machines.

Le problème de planification de processeurs parallèles dans un environnement de fabrication pour stocks à coûts d'installation dépendant des séquences, est examiné. Un nouvel algorithme formulant une série de sous-problèmes en nombres entiers 0-1 est proposé et comparé à une formulation antérieure (Dearing et Henderson 1982, 1984). Des parallèles entre les formulations des sous-problèmes et les réseaux généralisés sont examinés. L'efficacité et la qualité des solutions fournies sont testées au moyen de données antérieurement publiées pour un problème d'allocation de métier. La solution heuristique a été évaluée par rapport à la solution de programmation linéaire en nombres entiers optimale (ILP: Integer Linear Programming) et une approximation arrondie de programme linéaire (LP: Linear Program) pour la solution optimale à de nombreux problèmes d'échantillons. Les résultats obtenus montrent que l'heuristique est efficace, qu'elle apporte des solutions quasi-optimales aux problèmes de planification de la production et qu'elle nécessite une capacité informatique sensiblement inférieure par rapport aux approches LP et ILP pré-citées.

Es wird das Problem der Ablaufplanung von parallelen Prozessoren in einem auf Vorrat arbeitenden Betrieb erwogen, bei dem die Rüstkosten von der Arbeitsfolge abhängig sind. Ein neuer Algorithmus, der eine Reihe von ganzzahligen Null-Eins-Teilproblemen aufstellt, wird vorgeschlagen und mit einem früheren Rechenverfahren verglichen (Dearing and Henderson 1982, 1984). Parallelen zwischen der Teilproblemformulierung und verallgemeinerten Netzen werden besprochen. Anhand früher veröffentlichter Daten für die Zuteilung von Webstühlen werden der Wirkungsgrad und die Güte der zur Verfügung stehenden Lösungen untersucht. Die heuristische Lösung wird an einigen Musterproblemen mit der optimalen ganzzahligen linearen Programmierung (ILP) sowie einer abgerundeten Linearprogrammierungsannäherung (ALP) an die optimale Lösung verglichen und bewertet. Die Ergebnisse lassen darauf schließen, daß die heuristische Methode leistungsfähig ist, nahe am Optimum liegende Lösungen für Probleme der Fertigungsplanung liefert, und bedeutend weniger Rechnerkapazität erfordert als die ALP- und ILP-Lösungsansätze, über die früher berichtet worden ist.

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